1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

High stromal-to-tumor ratio defines a microenvironment that increases MTOR pathway dependence in bladder cancer; MTOR inhibition induces autophagy-driven cytotoxicity in TP53-wildtype tumors, whereas TP53-mutant tumors redirect autophagy toward survival, reducing therapeutic efficacy. Clinically, TSR-high, MTOR-dependent, TP53-wildtype bladder cancers will display improved overall survival under MTOR-targeted therapy compared with TSR-low or TP53-mutant tumors.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MTOR, TP53
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MTOR: mTOR-targeted therapy (everolimus-class) induce autophagy-dependent tumor cell death in some urothelial/bladder cancers but not others, and how does TP53 status — together with stromal context — determine whether autophagy is cytotoxic versus cytoprotective? (2) Tier 1 paths used: CORE PATH 1 (MTOR → everolimus → autophagy → cell death) provides the mechanistic drug–pathway axis; CORE PATH 2 (TP53 association with urothelial carcinoma) supplies a genetic modifier that plausibly alters cell-fate decisions downstream of autophagy.

TP53: TP53 status — together with stromal context — determine whether autophagy is cytotoxic versus cytoprotective? (2) Tier 1 paths used: CORE PATH 1 (MTOR → everolimus → autophagy → cell death) provides the mechanistic drug–pathway axis; CORE PATH 2 (TP53 association with urothelial carcinoma) supplies a genetic modifier that plausibly alters cell-fate decisions downstream of autophagy.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	EVEROLIMUS	Supporting modifier from knowledge graph
Secondary	AUTOPHAGY	Supporting modifier from knowledge graph

Secondary	METASTASIS	Supporting modifier from knowledge graph
Secondary	DIABETES	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In silico analyses of bladder cancer cohorts to correlate TSR scores with MTOR/TP53 status, autophagy gene signatures, and OS.

Prediction 2: In vitro models using bladder cancer lines with known TP53 status in a stromal-like microenvironment (e.g., fibroblast co-cultures or ECM-altered conditions) to test MTOR inhibitors, monitor autophagic flux (LC3-II, p

Prediction 3: , viability, and death, and determine dependence on autophagy by adding autophagy inhibitors.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EVEROLIMUS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- AUTOPHAGY: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.